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AutoStreamML: The Effective Way to Train Machine Learning Model Data Insights

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Abstract— AutoML is a no-code platform that enables easy development of machine learning models for people with limited coding expertise. Its primary focus is on providing the best user experience (UX) for uploading datasets, conducting exploratory data analysis (EDA), and training machine learning models through automation. The platform implements AutoML using TPOT, the user interface (UI) is built on Streamlit, with data pre-processing and model evaluation done using Scikit-learn. Some of the main advantages includes automating feature selection, feature engineering, preprocessing, and model tuning, which lessens the manual workload. Datasets can be uploaded, their structure examined, and automated reports generated using ydata profiling. The system can also manage data of various types dynamically such that numerical and categorical variables are processed seamlessly. Core features are finished, and work remains for integration with SQL and the RAG model. The objective is described as allowing natural language queries to be posed against the dataset, enabling effortless interaction with the data compared to the cumbersome complex SQL queries that need to be written. From the dataset, the highest-performing algorithm was automatically selected by the TPOT AutoML tool. The identified model, optimized through genetic programming, was nearly 90% accurate.

Keywords—AutoML, No-Code Machine Learning, TPOT, Streamlit, Data Preprocessing, Feature Selection, Exploratory Data Analysis.

I. INTRODUCTION

This no code AutoML platform simplifies machine learning by automating complex steps like tasks, classification, data preprocessing, and model selection. The platform does not require deep technical knowledge, enabling users even those from a non-technical background to construct and implement machine learning models with minimal effort.

AutoML powered by TPOT removes manual hyperparameter tuning, automatically selecting the best classification models, which greatly reduces development time. The integration of Pandas profiling improves the workflow by producing detailed automated reports, guaranteeing the data is properly organized before model training.

Through generative AI's active predictive capabilities and the platform's cost-saving integrations, it is a fitting option for Artificial Intelligence solutions for startups and small to medium enterprises looking to minimize expenditure.

Furthermore, the platform enables easy model export and subsequent deployment, which affords businesses the ability to utilize optimized models in practical scenarios directly. Upcoming developments comprise an SQL and RAG (Retrieval-Augmented Generation) model, which uses natural language with regular SQL to formulate accurate data commands and questions with object-oriented A.I.; thus, improving human-computer interaction simplifies vision AI-driven analytics.

The platform strives to improve accessibility to machine learning by reinforcing user efficiency alongside system proficiency, with the essence of versatility. Integrating AutoML and Athelike query-driven-AI makes it probable that non-technical insight-driven changes will be made by professional MBA holders onto systems.

Independently commissioned studies show that competition-grade accuracy alongside unparalleled reduction in time during model development is achieved with the supplied platform. The provided interface and the automated features widely improve productivity. The low entry barriers alongside attempts at streamlining intricate tasks increases the overall outcome.

Contribution: There is increasing need for machine learning platforms that make it easy to use for non-coders such as students and startups. Current tools have limited user interfaces and need one to know coding. This work adds value by providing automated model creation with TPOT, inbuilt data profiling, and scheduled Generative AI integration. Constraints are less flexibility for expert users and dealing with large or unstructured data.

II. LITERATURE REVIEW

Recent studies illustrate the increasing convergence of machine learning (ML), deep learning (DL), AutoML, and generative AI across multiple fields including financial risk analysis, healthcare, academia, and software development. Darji et al. introduced a novel framework using Retrieval-Augmented Generation (RAG)-based large language models to advance financial risk analysis, illustrating how generative AI can improve sophisticated financial decision-making [1].

Ved and Gupta also showcased the significance of massive data preprocessing and exploratory data analysis (EDA) for enhancing the accuracy of ML/DL models for house price estimation, and also underlined the significance of AutoML to ease model development for users who are not experts [2].

Concurrently, Lamb and Marimekala explored the ethical and cognitive aspects of using generative AI in academic settings, highlighting the sensitive balance between technological progress and ethical use of AI [3].

The relative efficacy of AutoML for various ML and DL algorithms was examined by Ferreira et al., and it showed that it can simplify intricate model selection and hyperparameter tuning tasks [4].

Wu et al. also expanded this idea further by incorporating AutoML into the training process of the data pipeline, improving overall model optimization [5].

In SMEs, Ahmad et al. have investigated the integration of generative AI in building supply chain resilience, especially in uncertain situations [6].

Huber and Kovalerchuk introduced a no-code platform (DV 2.0) for visual discovery of knowledge, bringing AI-assisted data analytics to the masses of users without coding skills [7].

The design of faster AutoML algorithms like TPOT-SH was proposed by Parmentier et al., solving scaling issues with big data [8].

Narasimhulu et al. investigated how containerization can streamline deployment procedures in DevOps, connecting software engineering and AI model deployment [9].

Kirmemis and Tekalp highlighted the pivotal role of training and test data in the performance of deep learning models in image restoration and super-resolution applications [10].

Dhyani et al. utilized generative AI to generate API documentation automatically, minimizing human effort and enhancing developer productivity [11].

Kumar et al. created interactive educational aid systems based on large language models with a goal towards personalizing academic support [12].

Saini et al. proposed DoMoBOT, which is an AI-driven bot that performs automated and interactive domain modeling with a new approach to software engineering practices [13].

In the medical field, Ashwath et al. introduced a web application for ML-driven cardiovascular risk prediction to augment preventive healthcare functions [14],

whereas T.B.M. et al. suggested TextVerse, a Streamlit application for sophisticated document analysis, reflecting the explanatory flexibility of generative AI to process PDF and image documents [15].

As a whole, these works demonstrate the exponential advancement of AI technologies and their diverse applications in various fields.

III. PROPOSED METHODOLOGY

This project integrates an AutoML platform with various programming languages, frameworks, and machine learning tools. The platform relies on Python for data processing and machine learning tasks while Streamlit is implemented to build a dynamic web application that enhances user interaction with AutoML operations. Furthermore, Pandas is utilized for CSV file retrieval and processing, and ydata-profiling constructs automated EDA reports, adding significant value to the dataset's understanding through insight generation.

In the machine learning and AutoML domain, TPOT plays a vital role in model selection, feature engineering, and hyperparameter tuning automation. Preprocessing, model training, and evaluation are supported by Scikit-learn with train_test_split, LabelEncoder, OneHotEncoder, and Pipelines which optimizes the workflow. User-uploaded datasets are stored and managed through seamless CSV File Handling.

To improve user interaction, their interface is constructed with Streamlit, enabling users to upload datasets, view EDA reports, and start the AutoML process with exceptional ease. The implementation of TPOT takes away the burden of hyperparameter tuning and feature selection, automating the search for the best- performing models with provided pipelines. Best pipelines are then saved as Python scripts, ensuring effortless reuse and deployment of the developed models by users (best_model_pipeline.py).

As for Ollama, it is planned to empower AI-driven insights, allowing users to make natural language queries to find answers for analyzed datasets. LangChain will provide the SQL needed for storing and retrieving the data, while RAG, integrated with the SQL-bridging LangChain, will handle the SQL queries. Moreover, data analysis will be more accessible with the integration of AI-powered insights through Streamlit Chat, greatly enhancing interaction with insights from the data. These NLP capabilities will greatly enhance user engagement with the platform, bringing more analytic functions within reach of non-technical users.

Lastly, for model deployment and export, TPOT Export is utilized to save the optimized pipeline as best model pipeline.py, in order to TPOT Export preserve extensibility and configurability. Users have the ability to upload datasets and download trained models using Streamlit File Uploader & Downloader interface. The adding of error reporting system.

This system simplifies the machine learning workflow by combining AutoML with user-friendly tools like Streamlit, Pandas, and TPOT. Model accuracy is evaluated using the formula:

$$\text{Accuracy}= TP + TN / TP + TN + FP +FN$$

The best-performing pipeline is exported as a Python script for easy reuse. Future plans include integrating Ollama, LangChain, and RAG to enable natural language interaction with the data.

strengthen the robustness from careful logging and alerting of the service in the case of failures.

This is a no-code machine learning platform designed for users with low technical skills for complex processes as a result of integrating AutoML, SQL RAG, and AI tools. The platform also incorporates Python libraries, Streamlit, and AI tools, guaranteeing its scalability, flexibility, and simplicity, and empowers users. Further along in the project life cycle, additional enhancements such as cloud hosting, API access, and more sophisticated visualization features will be studied in order to improve the overall experience for users and increase use case applicability in multiple fields.

Users can interact with their data without needing to write complicated algorithms to analyze it programmatically. In the sphere of machine learning, TPOT is instrumental in automating model and feature selection, as well as hyperparameter tuning, increasing efficiency and streamlining the model development process. Supporting the workflow are Scikit-learn's basic functions like train_test_split, LabelEncoder, OneHotEncoder, and Pipelines for standardized preprocessing and evaluation.

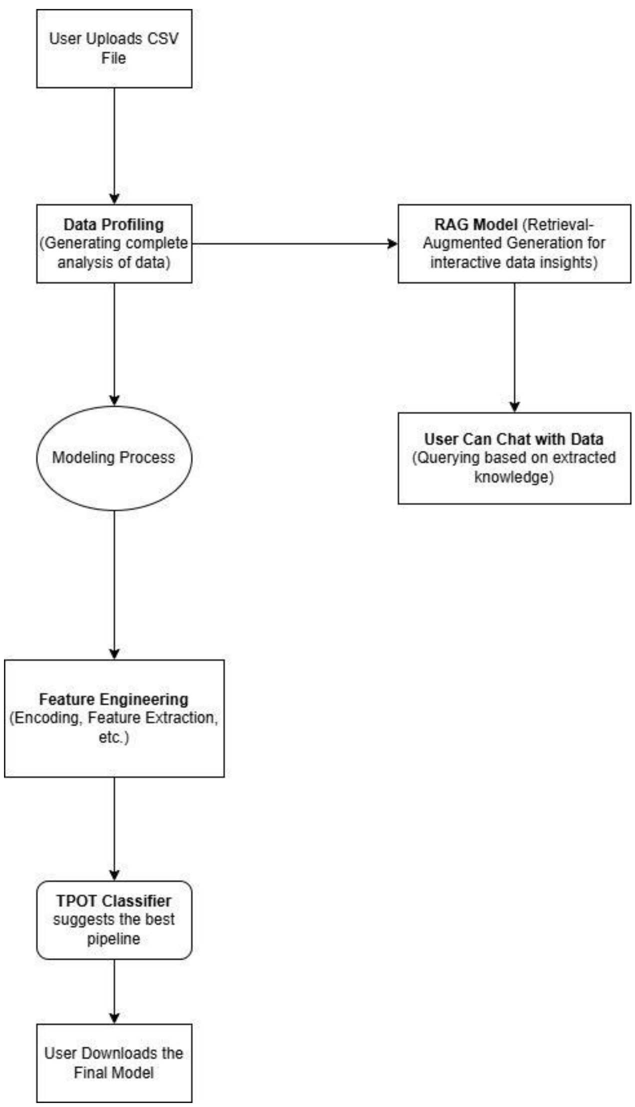


Fig.1.System Architecture

AutoML platform development requires multiple programming languages, frameworks, and machine learning technologies. The primary language is Python, which handles data processing as well as the machine learning tasks, while Streamlit and reimplementation of an interactive web application that automates AutoML processes and makes them more accessible for users. For data processing and analysis, Pandas performs operations on CSV files, ydata profiling completes exploratory data analysis (EDA), generating automated reports that summarize and provide important information about the dataset, assisting users in understanding their data better.

TPOT aids in feature engineering as well as training and optimizing machine learning models, automating model selection and hyper parameter tuning on AutoML design. Scikit-learn handles the rest of preprocessing, model training, and evaluation, providing train_test_split, LabelEncoder, One Hot Encoder, and Pipelines to streamline workflow of the operations. Users can upload datasets which are seamlessly managed with CSV File Handling and stored files ensuring effort-free access and effortless accessibility.

The platform encompasses a sleek interface developed with Streamlit that makes uploading datasets, obtaining EDA reports, and initiating AutoML tasks much simpler. Users may now sit back and relax as TPOT's integration handles the model selection by efficiently locating and executing the best models, thus removing the hassle of manual feature selection and hyperparameter tuning. After identifying the optimum pipe line it is exported as a Python script (best_model_pipeline.py). This guarantees that the users have the utmost convenience in reusing and deploying their models.

The plans are geared toward automation Ollama SQL RAG will enable powered insights serving AI based picture enabling users to pose natural language questions to structured datasets. SQL will be incorporated for data storage and retrieval while LangChain will serve as the SQL RAG model bridge enabling seamless SQL command execution. Furthermore, data analytics will be enhanced Streamlit Chat powered by AI will make insights conversational enabling the AI helps in analyzing the data. All of these features will unlock interactivity with data through plain spoken words helpful to none-technical users, leading to better intuitive understanding beyond complex advanced analytics for users.

As for the qualitative approach, it concentrates more on user experience and usability testing. The collection of feedback occurs via interviews and participatory observations where users interact with the Streamlit-based interface to evaluate the user-friendliness of dataset uploads, automated insights generation, and model deployment. Future modifications include the addition of Ollama SQL RAG which would allow users to issue natural language queries on datasets, thus further augmenting user accessibility and interaction. By integrating these approaches, this research aims to develop an efficient and user-friendly AutoML system designed for interacting with machine learning tasks.

The research method for this study comprised the following steps: A, First, data preprocessing was performed with Pandas where all missing values were dealt with, transformations were made, and categorical data was encoded. Automated exploratory data analysis (EDA) reports were generated using Ydata-profiling to ensure all aspects of the dataset were sufficiently covered. B, For model training and optimization, TPOT (Tree-Based Pipeline Optimization Tool) was used to automate the feature selection, hyperparameter tuning, and model selection within Scikit-learn pipelines structured ML workflows. Label Encoding and One-Hot Encoding were also used together with Imputation within the split dataset's train set. C, Development of the platform was completed in Streamlit and consisted of an interactive web interface.

insights can be retrieved through a conversation. Connecting LangChain was developed to bridge the gap between SQL databases and RAG models to increase the system's analytical functioning. E. Additionally, model export and deployment mechanisms were configured so that the best-performing pipelines could be captured and disseminated for further utilization. Active model evaluation using accuracy, F1-score, and confusion matrix performance metrics was implemented to give users insight into model effectiveness and provide visibility and improved decision-making ML automation governance.

This platform combines AutoML, SQL RAG, and AI insights which provides a tailored experience requiring zero coding, tailor-made for users with low technical skills. The extensive libraries provided by Python, combined with the user-friendly UI from Streamlit, and the AI-powered upgrades guarantee that the platform is easy to use, flexible, and maintainable. Enhanced user experience, broadened domain application, explorations of cloud integration, API, and advanced tools such as sophisticated visual aids will be added to the project as it matures.

AutoMLWorkflowwithAI-PoweredDataInsights User

Uploads CSV File

- The userprovidesadatasetinCSVformattoinitiateth e AutoML process.

DataProfiling

- Theplatformperformsautomatedexploratorydataanalysis (EDA) using tools like ydata-profiling, generating a comprehensive summary of the dataset.

RAGModel(Retrieval-AugmentedGeneration) *(Parallel Process)*

- TheextracteddatainsightsareprocessedusinganAI-powered RAG model to enable interactive data exploration.
- Thisallowsuserstoquerytheirdatasetusingnatural language.

UserCanChatwith Data

- UserscaninteractwiththedatasetviaanAIchatinterface, extracting insights based on the profiled data.

ModelingProcess

- ThemainAutoMLpipelinebeginsafterdataprofiling,leading to feature engineering.

FeatureEngineering

- Encoding and feature extraction techniques (e.g., Label Encoding,One-HotEncoding)areappliedtopreprocessthe data.

TPOTClassifier

- TPOT,anAutoMLtool,exploresdifferentpipelinesand suggests the best-performing model.

UserDownloadstheFinalModel

- OnceTPOTfinalizesthebestmodel,theusercandownloadit for deployment or further analysis.

IV. Results and Discussion

The Machine Learning workflow gets automated with the AutoML platform, enabling ease of access for users who do not possess advanced technical skills. The platform utilizes sophisticated tools and libraries, automating the entire procedure, including data analysis, model selection, and even model deployment. The integration of TPOT with the platform ensures that automated model selection along with hyperparameter tuning is performed, making sure that the best machine learning models are chosen with little human interaction. Traditional model experimentation is time and effort intensive which this approach eliminates while also enhancing the predictive performance of the models. Exploratory data analysis—enhanced by Panda's profiling—is another automated feature included with the framework which aids users in automating EDA tasks. Detailed reports containing data characteristics like missing values, correlational metrics, distributions, or presence of outliers can be prepared within no time. Such features are essential for gauging the data quality and preparing the data for model training. The user experience is made better through Streamlit which simplifies the handling, processing, visualization, and deployment of models, serving as a potent interfacing tool. Users, through the clear and well-designed UI, can interact with the platform by uploading datasets, gaining insights, model selection, and interpreting results, all without needing extensive coding knowledge. Moreover, realtime predictions and model interpretability is leveraged through SHAP (SHapley Additive Explanation) which provides unobstructed insight into how decisions are made.

The automated feature engineering to enhance model performance is achieved by transforming raw data into meaningful features with intelligence. Through these capabilities, the AutoML platform, identifiers and implements the algorithms and model workflows single handedly, augments the speed of the machine learning workflow and also democratizes advanced analytics engineering AI scaled technologies available to everyone.

The platform underwent testing with a variety of datasets, proving its potential in determining the ideal models and hyperparameter settings in a timely manner. The manual workload attributed to model tinkering was greatly alleviated by implementing TPOT's genetic algorithm, yielding competitive outcomes for classification and regression tasks. The Streamlit interface simplifies the AutoML process, allowing users to upload datasets, visualize key insights, and deploy models without writing code. Natural language access to dataset analysis queries makes upcoming features like Ollama SQL RAG for AI-powered insight analysis increase accessibility greatly.

Some of the challenges to be dealt with include accommodating structured and unstructured data, integrating profiling for big data, and increasing the efficiency of the pipeline. Also, real-time inference and model interpretability are still lagging behind others in advancement. Solving these problems will strengthen and increase the scalability of the platform.

Ensuring bias mitigation and fairness remains as another key gap because of ethical AI obligations in real-life implementations. While AutoML simplifies model development, it's just as critical that such models are not biased because of the training data's inherent biases. The platform will adopt fairness metrics moving forward, along with SHAP and LIME, which will unmask models' decision-making processes to enhance trust in ML. These techniques will enable users to analyze the predictive logic, thereby nurturing trust and responsibility in machine learning systems.

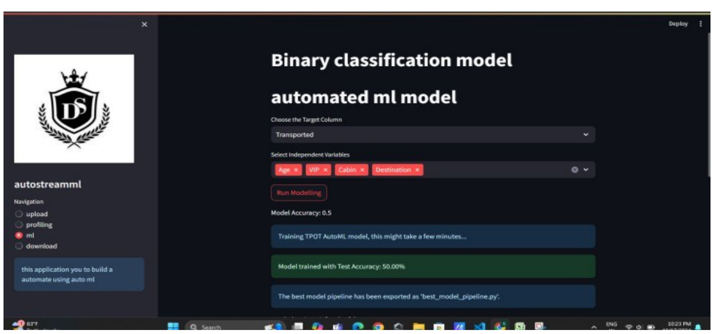
Moreover, automated feature selection and engineering will become primary areas of focus. Ameliorations in feature TPOT's feature selection will further optimize accuracy and model efficiency TPOT already does on some level. The platform can apply domain-specific feature engineering to various sectors like healthcare, finance, and e-commerce to enhance model performance.

The results, as stated previously, illustrate the platform's advantage in the democratization machine learning by performing intricate tasks while automating them and retaining flexibility and efficiency. Efforts will be directed towards increasing explainability, bettering handling large amounts of data, and other AI add-ons toward a more complete experience for users of AutoML.
The platform has also excelled across multiple industries like marketing, finance, healthcare because of its versatility. It helps non-expert users enable teams in harnessing the capabilities of machine learning without having to rely on data science professionals, which is why it is so valuable. As a result of the automation of model selection, hyperparameter optimization, and feature engineering, the overall advancement achived is with respect to the accuracy and relevance of the created models as well as developed time.

Having the possibility of deploying models straight from the platform increases the speed of model rollouts and enhances agile, data- driven organizational decision making. It will strengthen innovation and efficiency across industries that seek to incorporate AI into their workflows.

The project, at this stage, goes far beyond the technical boundaries, because it has created some remarkable cross disciplinary innovation with rest of the team. A platform design was created by combining modern data science, software engineering, and UX design that fulfills the existing market conditions and anticipates future modifications, emerging innovations. This work has shown how vital continuous and adaptive change is, thus leading to more research around cloud solutions, security, and AI ethics. Because of this, the project can now be further developed into a sophisticated system able to tackle complex predictive tasks across different industries.

The results of the research data will be presented as follows, in the Figure below:



Fg2.dependent and independent variables selection.

AutoStreamML is a sophisticated web application that seeks to streamline the steps involved in executing machine learning projects with the help of AutoML features. It allows users to seamlessly upload datasets, profile them, train models and download optimized pipelines. The application simplifies the operation of machine learning systems, especially for binary classification problems, by using the machine learning algorithms available in Scikit-learn. The principal features consist of reading CSV files containing datasets, splitting datasets into training and testing data, and performing optimized classification with SGDClassifier. This model is trained on specific hyperparameters that give the best results. The interface has a navigation index for easy profiling and data uploading and for machine learning execution and model downloading. Having been trained, the model is easy to export into other environments for use which makes the application better placed in performing automation of ML tasks. The manual effort of selecting a model or tuning the model using hyperparameters is done with less effort as a result of the techniques applied which improves work done using machine learning.

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Overview | Alerts 21 | Reproduction

Dataset statistics	
Number of variables	14
Number of observations	92
Missing cells	22
Missing cells (%)	1.7%
Duplicate rows	0
Duplicate rows (%)	0.0%
Total size in memory	8.9 KiB
Average record size in memory	99.4 B

Variable types	
Text	3
Categorical	2
Boolean	3
Numeric	6

Fig3.Table

This image exhibits a data profiling report, probably produced by a profiler of Pandas, featured in the Autostreamml ecosystem. It illustrates a summary of data containing 14 variables and 92 observations. The missing cells were determined to be 22, which is 1.7% of the data. There are no duplicate rows. The report also shows the memory footprint of the dataset (8.9 KiB). It also classifies the variables into different types as text, categorical, boolean, and number columns. The report alerts21 suggests relevant alerts that draw attention within the dataset.

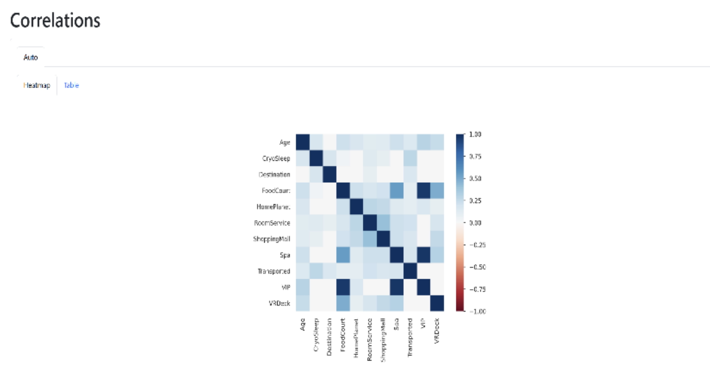

Fig.4.Corelation

This image from an AutoStreamML project shows a correlation heatmap. The aim of this map is to summarize the relationships among different variables in the dataset. The variable's relationships can be fully analyzed using the heatmap as coloration demonstrates clearly whether there is a positive or negative value between the two variables using a gradient, -1.00 representing negative and 1.00 positive. They are listed on both axes which make it easy to find both waist and fist pair correlation. The linear relationships among various features within the dataset, with color intensity and hue representing the strength and direction of correlation, dark blue being strong positive correlation and grays toward white meaning weak or no correlation. Inspection of the heatmap provides feature interaction insight, for example, high positive feature correlations between spending features such as Spa, RoomService, and VRDeck, and indicates relative absence of high linear correlation between features such as Age, CryoSleep, or Destination and others, offering critical information for feature understanding, choosing, and possible engineering inside the automated machine learning cycle.

V. CONCLUSION AND FUTURE WORK

The progress made on this AutoML platform automates an end-to-end machine learning workflow by providing a friendly interface that requires no coding knowledge regardless of a user's experience. BHandling data from TPOT for automated model selection, pandas profiling for exploratory data analysis, and a Streamlit-powered GUI greatly improves dataset preprocessing and model deployment. The addition of Ollama SQL RAG (for later integration) strengthens feature accessibility by allowing natural language AI queries for data insights. This research, through a blend of qualitative user feedback and quantitative performance assessment, illustrates how effectively AutoML minimizes manual processes and expedites model development. This not only enhances operational efficiency but also makes advanced automated machine learning services available to startup and small and medium-sized enterprises along with their non-technical staff. Further versions of the product will incorporate additional data type support and improve model construal to aid user decision making. Beyond these underscored core advantages, the project showcases the impact of eradicating barriers to the world of machine learning. The algorithm's innovation promotes informed decision-making by providing easy access to advanced ML tools. Considering the future, the following versions will implement self-explanatory modules and instant reasoning to enhance understanding into model's decision-making processes. This will provide users the ability to comprehend and have faith in the system's outputs without the need to code, which is an approach geared towards ease of use. Our goal is to strengthen the platform as a powerful, easy to use, and advanced framework that serves a wide audience by integrating new AI developments and optimizing the systems performance, thereby revolutionizing machine learning.

REFERENCES

[1] A. Darji, F. Kheni, D. Chodvadia, P. Goel, D. Garg and B. Patel, "Enhancing Financial Risk Analysis using RAG-based Large Language Models," 2024 3rd International Conference on Automation, Computing and Renewable Systems (ICACRS), Pudukkottai, India, 2024, pp. 754-760, doi: 10.1109/ICACRS62842.2024.10841711.

[2] A. R. Ved and A. Gupta, "Extensive Data Preprocessing and Exploratory Data Analysis for House Price Prediction using Machine Learning and Deep Learning with an Introduction to AutoML Technologies," 2024 IEEE 3rd World Conference on AppliedIntelligence and Computing (AIC), Gwalior, India, 2024, pp. 711-721.

[3] D. Marimekala and J. Lamb, "Impact of Generative AI Adoption in Academia and How it Influences Ethics, Cognitive Thinking and AI Singularity," 2024 IEEE Integrated STEM Education Conference (ISEC), Princeton, NJ, USA, 2024, pp. 1-4, doi: 10.1109/ISEC61299.2024.10664756.

[4] Ferreira, A. Pilastri, C. M. Martins, P. M. Pires and P. Cortez, "A Comparison of AutoML Tools for Machine Learning, Deep Learning and XGBoost," 2021 International Joint Conference on Neural Networks (IJCNN), Shenzhen, China, 2021, pp. 1-8, doi: 10.1109/IJCNN52387.2021.9534091.

[5] J. Wu, H. Wang, C. Ni, C. Zhang and W. Lu, "Data Pipeline Training: Integrating AutoML to Optimize the Data Flow of Machine Learning Models," 2024 7th International Conference on Advanced Algorithms and ControlEngineering(ICAACE), Shanghai, China,2024,

[6] K. Ahmad, A. Rozhok and R. Revetria, "Supply Chain Resilience in SMEs: Integration of Generative AI in Decision-Making Framework,"2024 International Conference on Machine Intelligenceand Smart Innovation (ICMISI), Alexandria, Egypt, 2024, pp. 295-299.

[7] L. Huber and B. Kovalerchuk, "No-Code Platform for Visual Knowledge Discovering in General Line Coordinates: DV 2.0," 202327th International Conference Information Visualisation (IV), Tampere, Finland, 2023, pp. 316-322, doi: 10.1109/IV60283.2023.00060.

[8] L. Parmentier, O. Nicol, L. Jourdan and M. -E. Kessaci, "TPOT-SH:AFasterOptimization AlgorithmtoSolvetheAutoMLProblemon Large Datasets," 2019 IEEE 31st International Conference on Tools with Artificial Intelligence (ICTAI), Portland, OR, USA, 2019, pp. 471-478, doi: 10.1109/ICTAI.2019.00072.

[9] M. Narasimhulu, D. V. Mounika, P. Varshini, A. K and T. R. K. Rao, "Investigating the Impact of Containerization on the Deployment Process in DevOps," 2023 2nd International Conference on Edge Computing and Applications (ICECAA), Namakkal, India, 2023, pp. 679-685, doi: 10.1109/ICECAA58104.2023.10212240.

[10] O. Kirmemis and A. M. Tekalp, "Effect of Training and Test Datasets on Image Restoration and Super-Resolution by Deep Learning," 2018 26th European Signal Processing Conference (EUSIPCO), Rome, Italy, 2018, pp. 514-518, doi: 10.23919/EUSIPCO.2018.8552961.

[11] P. Dhyani, S. Nautiyal, A. Negi, S. Dhyani and P. Chaudhary, "Automated API Docs Generator using Generative AI," 2024 IEEE International Students' Conference on Electrical, Electronics and Computer Science (SCEECS), Bhopal, India, 2024, pp. 1-6, doi: 10.1109/SCEECS61402.2024.10482119.

[12] P. Kumar, H. M and H. V, "Development of Interactive Assistance for Academic Preparation Using Large Models Language," 2025 International Conference on Computational, Communication and Information Technology (ICCCIT), Indore, India, 2025, pp. 265-269, doi: 10.1109/ICCCIT62592.2025.10928137.

[13] R. Saini, G. Mussbacher, J. L. C. Guo and J. Kienzle, "DoMoBOT: An AI-Empowered Bot for Automated and Interactive Domain Modelling," 2021 ACM/IEEE International Conference on Model Driven Engineering Languages and Systems Companion (MODELS-C), Fukuoka, Japan, 2021, pp. 595-599, doi: 10.1109/MODELS-C53483.2021.00090.

[14] S.Ashwath,G.Gurukishore,A.MaheedharandN.EdnaElizabeth, "Enhanced Cardiovascular Risk Prediction Using ML Powered Web Application,"202411thInternationalConferenceonSignalProcessing and Integrated Networks(SPIN),Noida,India,2024,pp.199-204,doi: 10.1109/SPIN60856.2024.10512336.

[15]T.B.M,R.H.V,R.S,R.SalamandM.Pai,"TextVerse:AStreamlit Web Application for Advanced Analysis of PDF and Image Files with and without Language Models," 2024 Asia Pacific Conference on InnovationinTechnology(APCIT),MYSORE,India,2024,pp.1-6,doi: 10.1109/APCIT62007.2024.10673559.